

Mason Drumwright

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EDUCATION

University of Colorado Boulder – College of Engineering

Anticipated: May 2027

B.S. in Aerospace Engineering

Minor in Engineering Management

- **Relevant Courses:** Aircraft Dynamics, Orbital Mechanics and Attitude Dynamics, Electronics and Communications, Experimental and Computational Methods, Aerospace Vehicle Design, Thermodynamics, Aerodynamics, Structures

EXPERIENCE

Research Assistant

Boulder, CO

Autonomous Vehicle Systems Lab

05/26 - Present

- Extending spacecraft dynamics to support the branching dynamics of rigid body subcomponents relative to the spacecraft hub by deriving the general translational and rotational equations of motion for complex spinning and translating systems
- Implementing decoupling dynamics in the open-source Basilisk astrodynamics simulation framework, expanding simulation capability for complex multibody spacecraft configurations
- Investigating MuJoCo-Basilisk framework integration to enable simulation of systems supporting flexible multibody dynamics models for future mission design

CU in Space – Competitive Rocketry Club

Boulder, CO

Guidance, Navigation, and Control Team

09/25 - Present

- Contributed to a system-level design of a hybrid rocket targeting a precise apogee of 10,000 ft using active airbrake control to increase drag and reduce velocity
- Designed state estimation algorithms to track the state of the rocket during ascent. Using attitude estimation based on accelerometer and gyroscope sensor fusion and an Extended Kalman Filter to estimate the full state
- Developed a Model Predictive control system using MATLAB and C++ that extends airbrakes based on predicted states from EKF by performing sensitivity analysis to understand how the rocket behaves with different control parameters/gains.
- Coordinated with airbrakes and avionics teams to integrate the flight computer's control algorithms with our airbrake motor drivers and other sensors on the rocket, using a shared Git repository with C++

RELEVANT PROJECTS

Fixed Wing UAV Build

Boulder, CO

Personal Project

4/26 – Present

- Simulated custom fixed-wing UAV geometries in MATLAB by implementing stability derivatives to validate a design by analyzing the aircraft's open and closed loop responses to perturbations
- Programmed a custom flight controller on a Teensy 4.1 in C++ that performs PID control based on a 15-state Extended Kalman Filter that fuses a 9-DOF IMU, GPS, and barometer to perform autonomous flight
- Developed guidance algorithms that use cascading PID loops by reading the EKF state to perform autonomous maneuvers including climb, cruise, and loiter.

University of Colorado Boulder

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Vehicle Design Lab

01/25 – 05/25

- Team project focused on the design and development of an aerospace vehicle capable of withstanding boost-ascent forces, executing a controlled pitch-over maneuver, and achieving stable gliding flight
- Performed analysis on equations of motion using numerical integration techniques (MATLAB) to complete trade studies, including trade-offs between increased wingspan and weight, lift, and drag for design improvement
- Programmed flight simulations in MATLAB based on different design characteristics in order to predict system performance and optimize the lift-to-drag ratio

Structural Beam FEM

Boulder, CO

Structures Lab

1/26 – 2/26

- Predicted beam deflection due to point loading with various weight forces by deriving expressions for mid-span deflection based on beam geometry and load forces, in order to estimate the predicted results of our simulations
- Created a FEM model of the beam in ANSYS and compared the simulated results of the model with our predicted deflection estimates, finding that they were very similar

SKILLS/ADDITIONAL INFORMATION

- Technical Skills: MATLAB, Python, C/C++, Onshape, Control Systems, ANSYS, SolidWorks
- Interests: Interplanetary missions in spaceflight, Aircraft/Spacecraft GNC systems, Skiing, Hiking, Soccer